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Multimodal integration reveals complementary imaging and genetic predictors of coronary artery disease risk

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github.com/Devanshpandey/Multimodal-integration-for-coronary-artery-disease-risk-prediction

Cardiac MRI embeddings and polygenic risk capture independent CAD risk — combining them identifies a subgroup with ≈23× the cohort event rate.

Colour key 🟧 imaging (cardiac MRI) 🟧 extra-cardiac imaging 🟦 genetics (PRS) ⬜ clinical / guideline scores ⬛ combined model

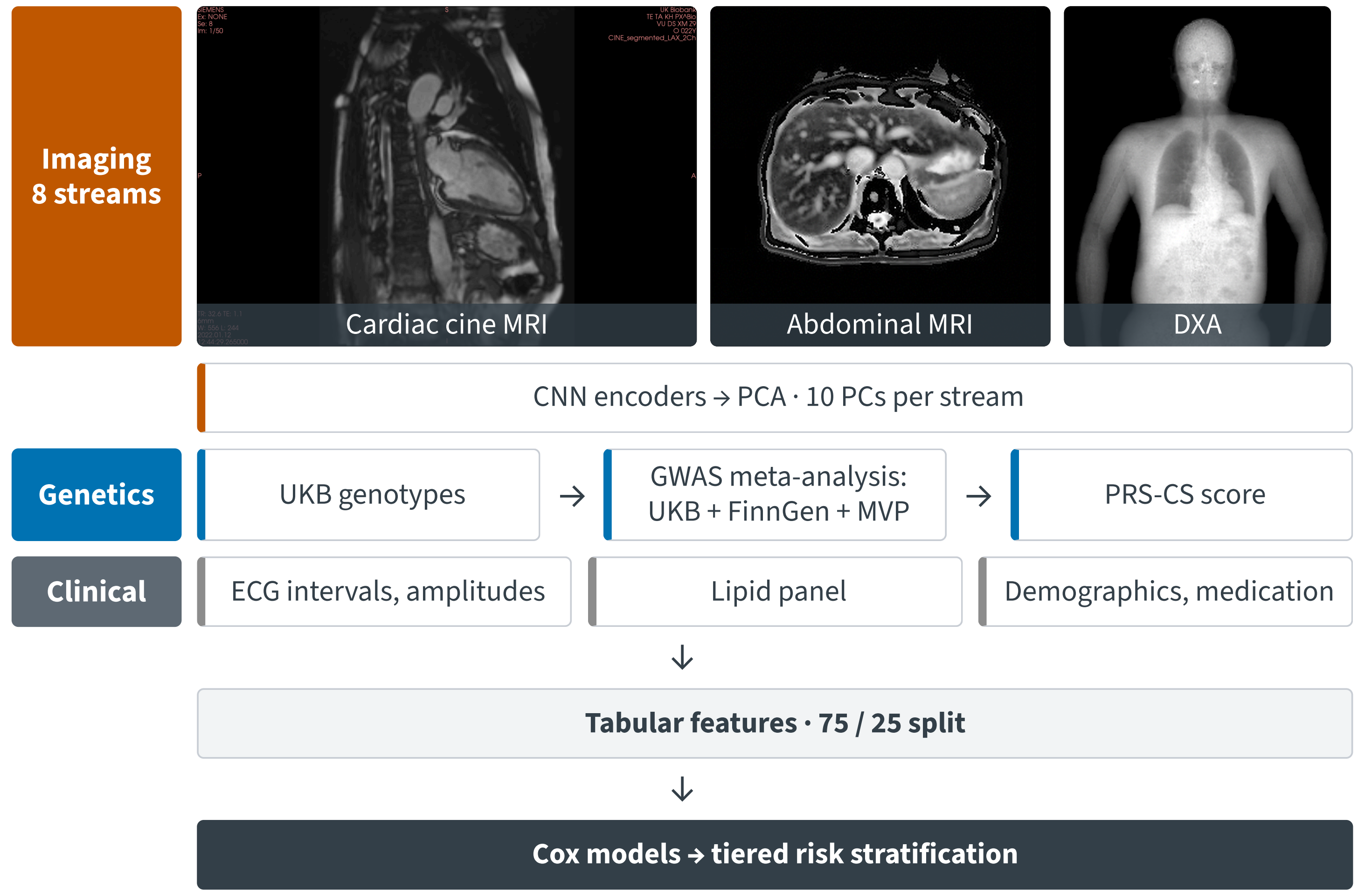
WHY & HOW

UK Biobank imaging cohort, three data domains, one model

Background & questions

- Leading cause of death; ~50% of CAD first presents as a heart attack.
- Clinical scores omit genetics and imaging; how much imaging and genetic risk overlap was unknown.

1. Which imaging modalities add to clinical, metabolic, ECG and genetic factors?
2. Overlap between imaging-derived and polygenic risk
3. Sequential integration: discrimination, calibration, stratification



CNN encoders → PCA · 10 PCs per stream

Genetics: UKB genotypes → GWAS meta-analysis: UKB + FinnGen + MVP → PRS-CS score

Clinical: ECG intervals, amplitudes · Lipid panel · Demographics, medication

↓

Tabular features · 75 / 25 split

↓

Cox models → tiered risk stratification

Cardiac MRI does not image coronary atherosclerosis; its signal reflects remodeling, vascular stiffness and cardiometabolic burden.

Outcome = clinically recognized CAD (MI, angina, revascularization); prevalent cases excluded.

42,351 UK Biobank imaging participants

677 incident CAD (1.6%)

4.6 yr median follow-up

211,755 cardiac cine MRI series

127,053 2D images (liver, pancreas, DXA)

10,597 held-out test set (178 cases)

Methods in brief

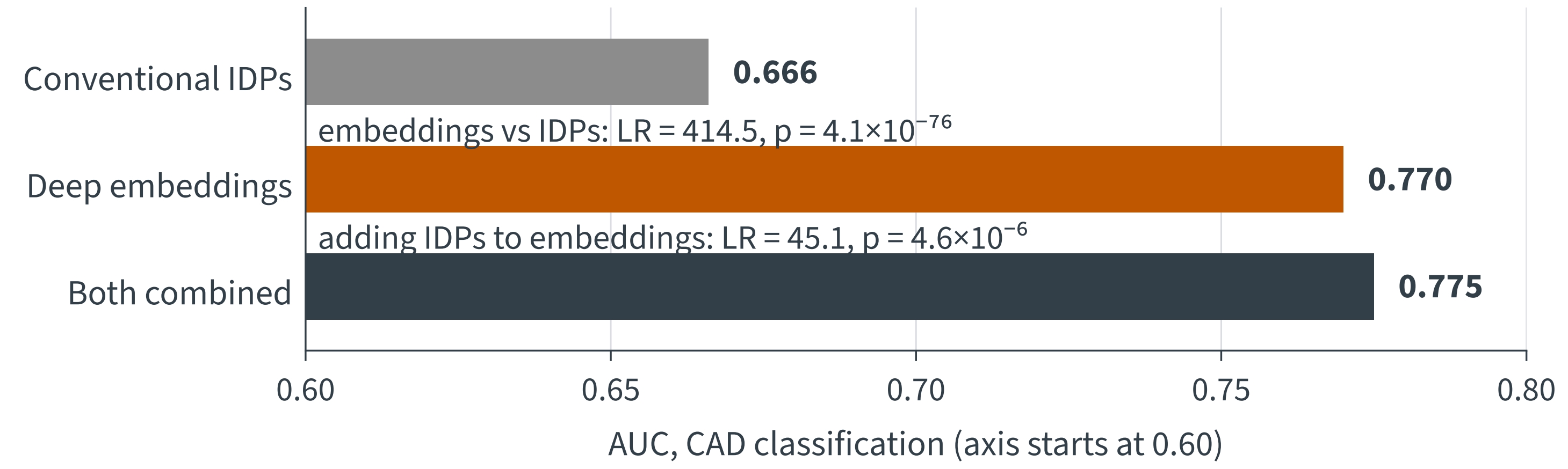
- Encoders fine-tuned on balanced case–control training data only.
- Embeddings → PCA, 10 PCs per stream (80 total).
- PRS-CS, European meta-GWAS: OR 1.87 per SD, AUC 0.766 (with age, sex, 20 PCs).
- XGBoost + nested Cox; imaging risk score = cross-fitted ridge Cox.

IMAGING REPRESENTATIONS

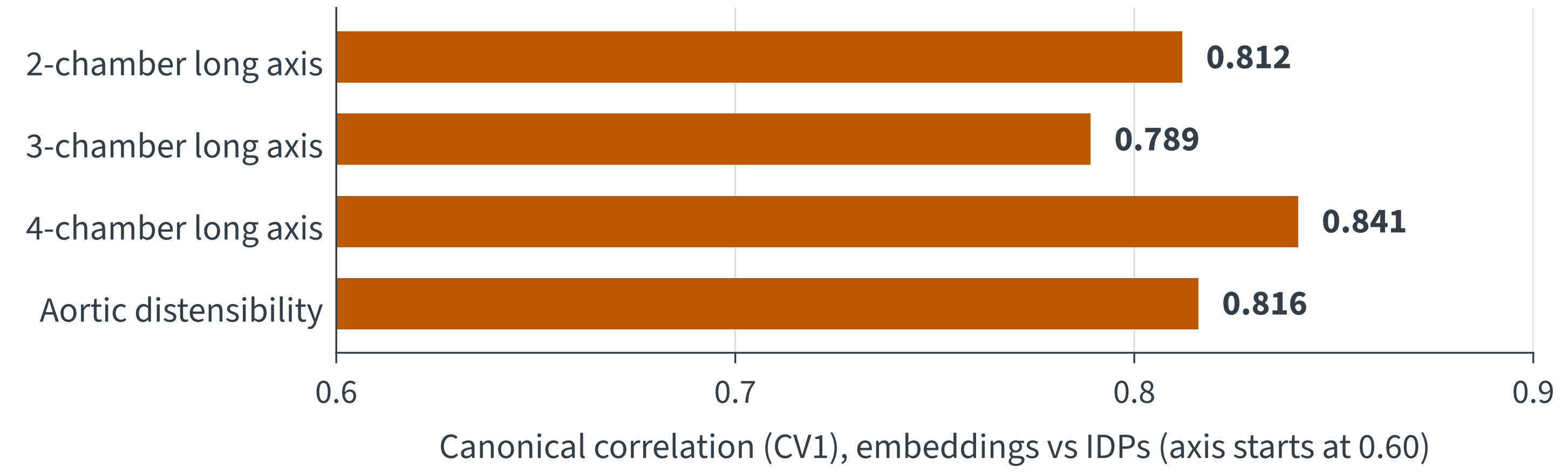
Deep embeddings beat handcrafted cardiac phenotypes

0.770 vs 0.666

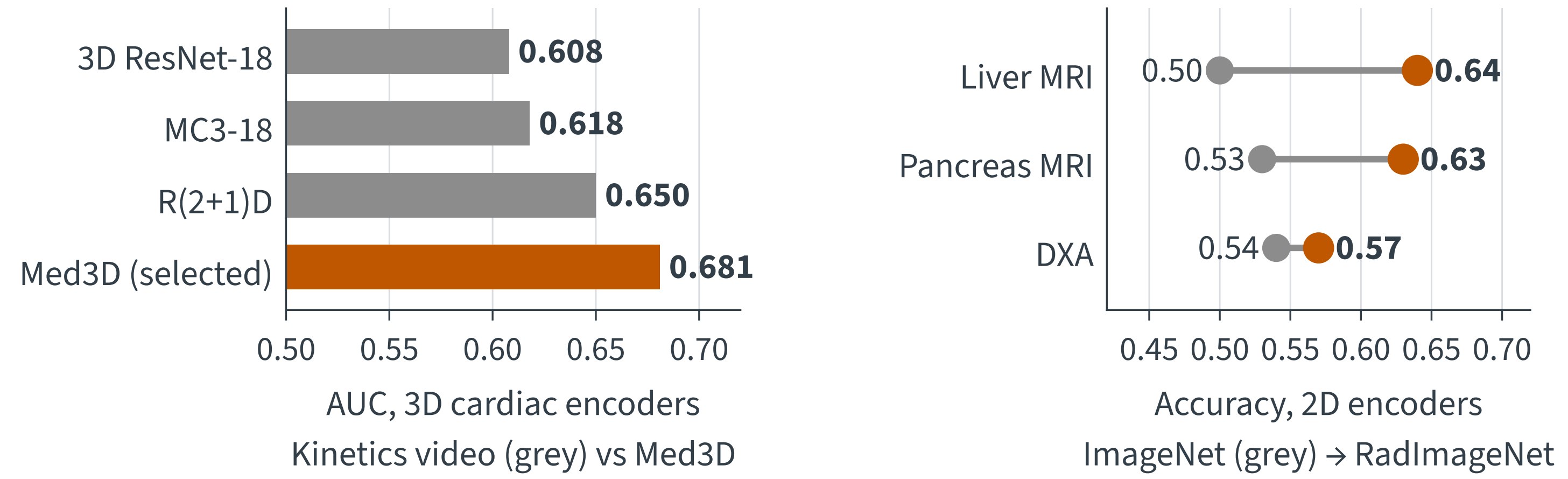
AUC, deep embeddings vs conventional imaging-derived phenotypes (IDPs); both combined 0.775



Embeddings outperform IDPs; adding IDPs barely helps (logistic models; performance plateaus at ~5 PCs per modality; Fig. 2a,b).



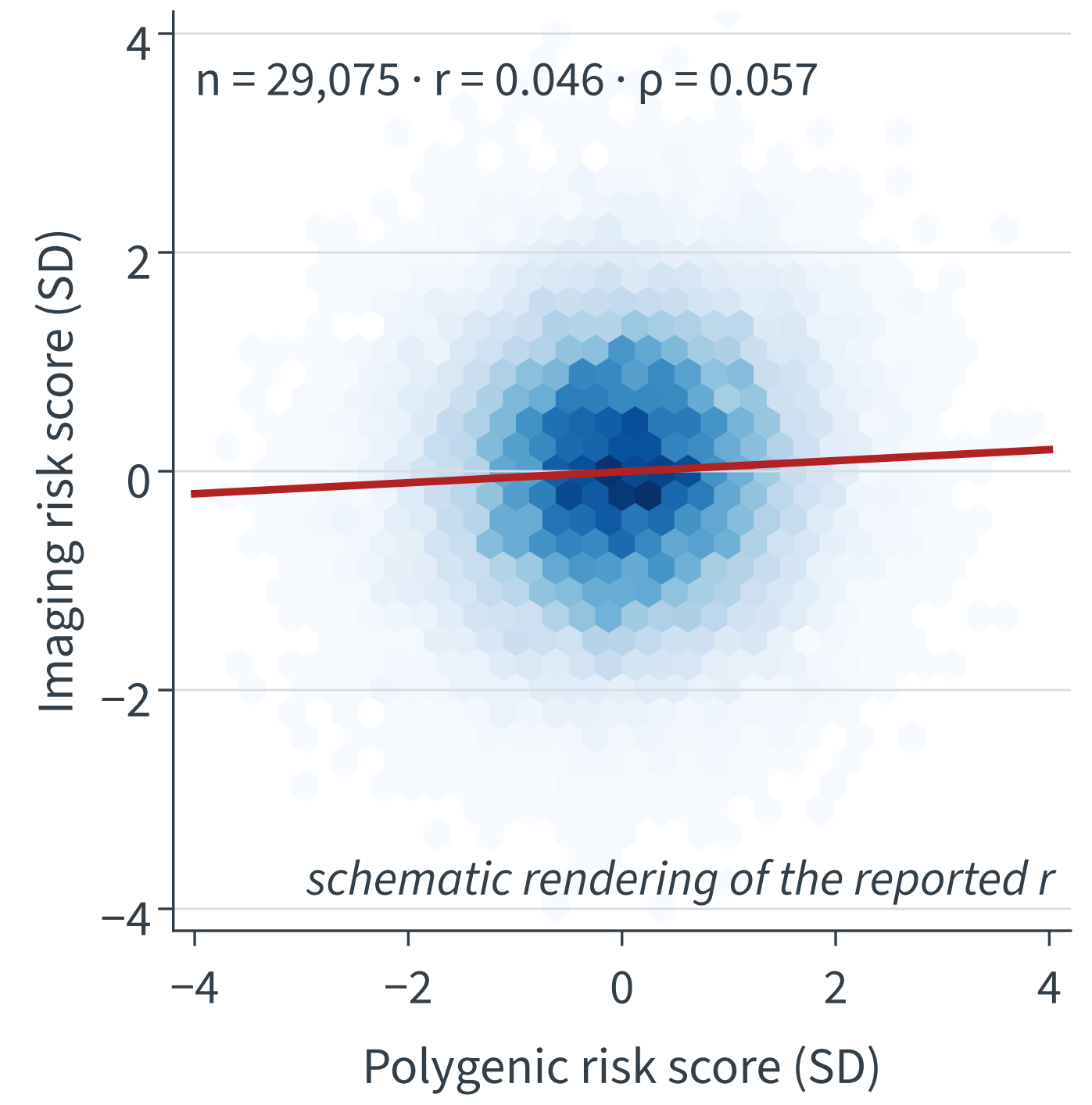
Embeddings encode physiology: CV1 loads on stroke volume, ejection fraction, myocardial mass and wall thickness (Fig. 2c).



Medical pretraining wins: Med3D for the cardiac 3D encoders, RadImageNet for liver, pancreas and DXA (axes from 0.50 / 0.45; Supp. Fig. 3b, 4b).

IMAGING VS GENETICS

Imaging and genetic risk are nearly independent



r ≈ 0.05

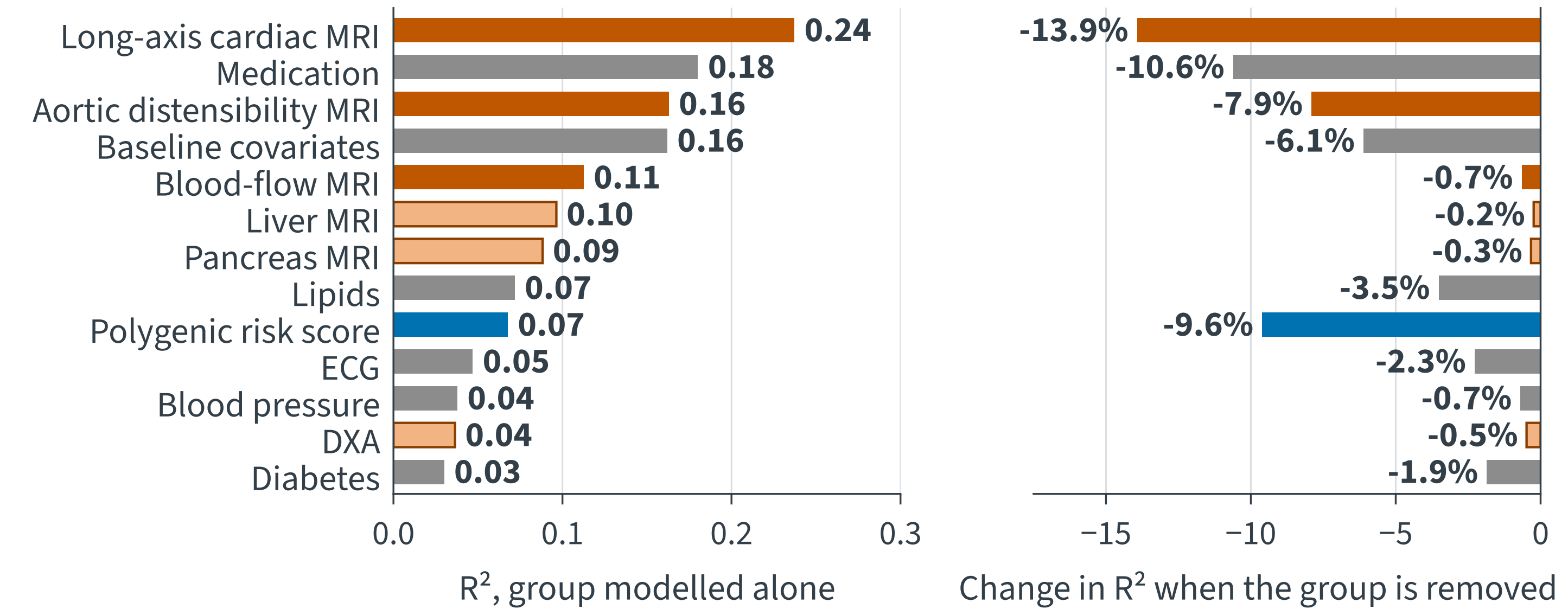
Pearson r, imaging vs polygenic risk score ($\rho = 0.057$); $R^2 < 1\%$ shared — the two overlap very little ($n = 29{,}075$; Fig. 3c–e).

Partial R^2 of imaging risk explained by

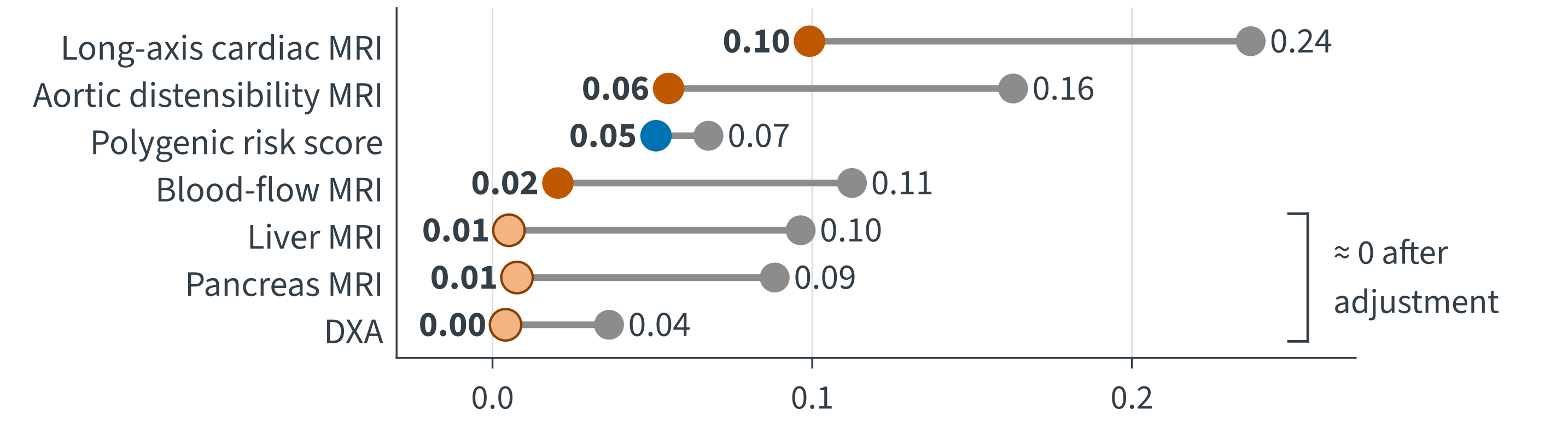
0.15% genetics (PRS)

38.4% demographics

14.4% clinical factors



Cardiac imaging, medication and PRS are each non-redundant. Left: R^2 of each group alone; right: R^2 drop when the group is removed from the full model (Fig. 3a,b).



Unique variance (R^2): grey = alone · coloured = after baseline adjustment

Only long-axis, aortic distensibility MRI and PRS retain unique variance after clinical adjustment (Fig. 3f).

INTEGRATION & STRATIFICATION

Sequential integration finds very-high-risk people

36.8%

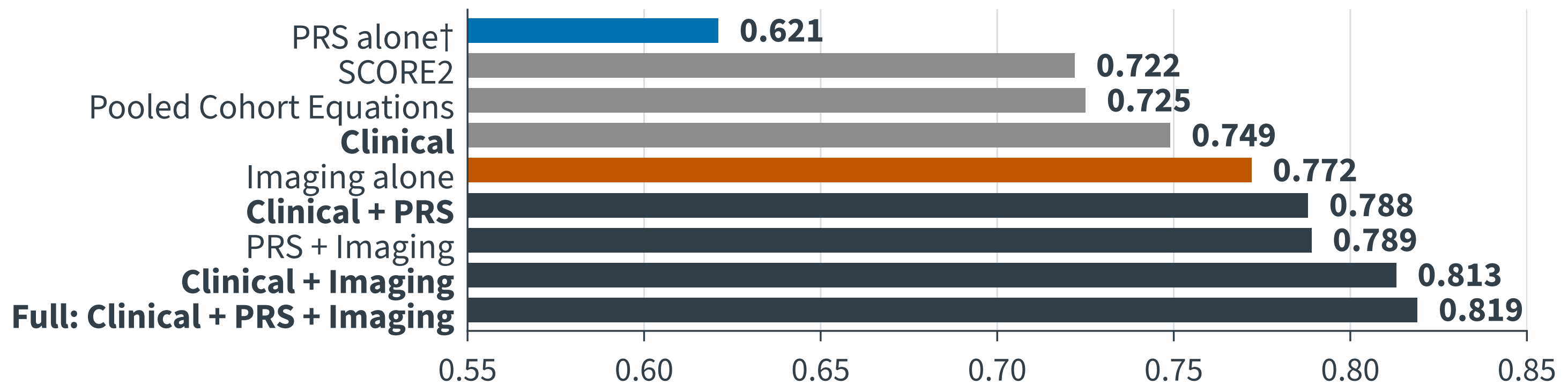
5-yr CAD incidence, top quintile of both imaging and genetic risk within Tier 3; cohort 1.6%

0.634 continuous NRI vs clinical (95% CI 0.534–0.734)

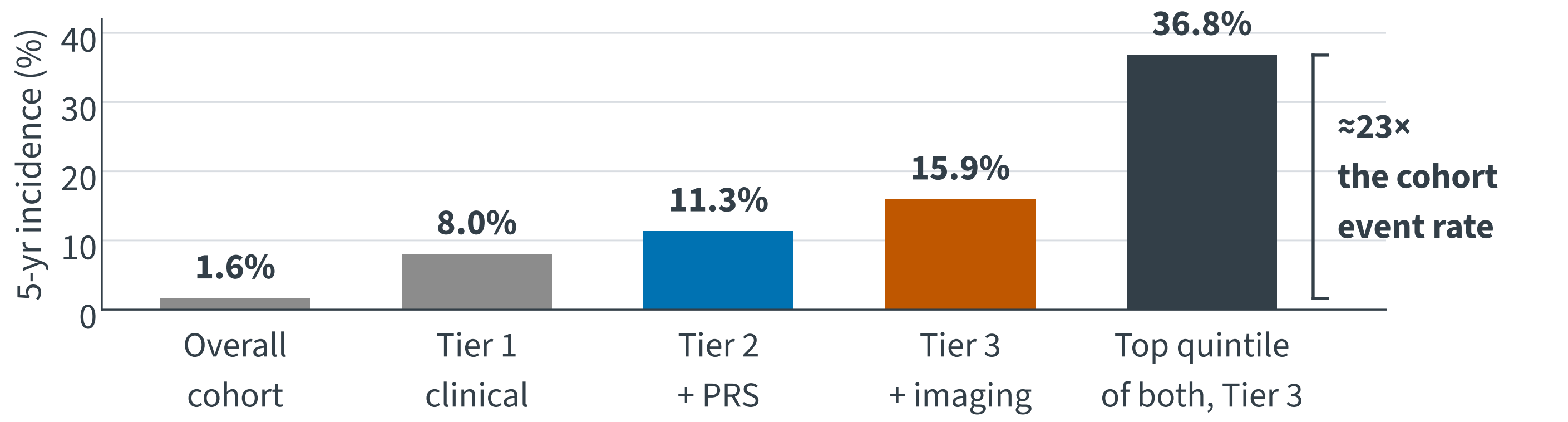
9.5% prediction accuracy (IPA), full model; clinical 5.5%

86.8 yr median heart age, Tier 3; cohort 55.3 yr

0.99–1.08 calibration slopes, nested models



Each layer adds discrimination; imaging is among the strongest independent predictors (HR ≈ 2.1 per SD). Held-out C-index; axis from 0.55; bold = nested. HR per SD, full model: imaging ≈ 2.1, PRS ≈ 1.7, demographics ≈ 1.4, lipids ≈ 1.1. †PRS alone = score without age/sex; AUC 0.766 (column 1) includes age, sex, 20 PCs.



Sequential tiers enrich risk far beyond clinical stratification (5-yr incidence, held-out set). Leave-one-centre-out: AUC 0.785–0.822, C-index 0.73–0.76; discrimination is stable across the four UK Biobank imaging centres (site-level robustness, not external validation).

Conclusions

- Imaging and polygenic risk are nearly uncorrelated: complementary axes.
- Long-axis cine and aortic distensibility MRI carry the incremental signal; blood-flow, liver, pancreas and DXA add little once clinical variables are included.
- Clinical → PRS → imaging integration isolates a 36.8% 5-yr-risk subgroup.
- A framework for complementary risk — not yet a clinical screening tool.

Limitations

- No CAC/CTA in UK Biobank: no calcium-score benchmark.
- European-ancestry, complete-case, healthier-than-general cohort; covariates measured at imaging, so part of imaging's gain may reflect temporal proximity.
- Not externally validated; observational — prospective trials needed.

Data & code. UK Biobank application 65439. Code at github.com/Devanshpandey/Multimodal-integration-for-coronary-artery-disease-risk-prediction (QR above). **Ethics.** UK Biobank participants provided informed consent; study conducted under UK Biobank approval. **Genetic disclosure.** The genetic component is a genome-wide polygenic score (PRS-CS, HapMap3 variants) from a European meta-analysis of non-imaged UKB + FinnGen + MVP (229,724 cases / 1,024,752 controls); no single gene is claimed. GWAS summary statistics and PRS-CS weights are deposited in the repository. **Acknowledgements.** Computing via the TACC Director's Discretionary Fund (Corral, Lonestar6, NVIDIA A100). D.P.: Outstanding Graduate Research Fellowship, UT Austin Graduate School. V.M.N.: Allen Discovery Center program (Paul G. Allen Frontiers Group) and a Good Systems Fellowship for Ethical AI, UT Austin. **Competing interests.** C.A.T. founded HeartFlow, Inc. and serves on the boards of HeartFlow, Inc. and Ebenbuild GmbH. Other authors declare none. **Key references.** UK Biobank: Sudlow 2015 *PLoS Med* · PRS-CS: Ge 2019 *Nat Commun* · GPSmult: Patel 2023 *Nat Med* · Med3D: Chen 2019 *arXiv* · RadImageNet: Mei 2022 *Radiol AI* · REGENIE: Mbatchou 2021 *Nat Genet* · SCORE2 2021 *Eur Heart J* · PCE: Goff 2014 *Circulation*. **Outcome definition.** Incident CAD = ICD-10 I20–I25, ICD-9 410/412/414, OPCS-4 K40–K50, K75; prevalent cases excluded; follow-up from the first imaging visit. **[Previously published: JOURNAL, DATE — remove if not applicable]**